

Mathematics A

Unit 1 • Chapter OL-T31 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

4 classified items • 23 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T31 • Expertise Q16+ • 4 items • 23 marks

Chapter	Items	Marks	Starts
01. OL-T31 Coordinate Geometry	4	23	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Coordinate Geometry

Unit 1 • Expertise • 23 marks • 4 item(s)

June 2022 | 4MA1-2H Q22 | 7 marks

Kite or symmetric vertex coordinates

January 2020 | 4MA1-1H Q22 | 6 marks

Coordinates from isosceles and gradient constraints

January 2021 | 4MA1-2H Q18 | 4 marks

Rectangle or parallelogram missing vertex

January 2023 | 1H Q23 | 6 marks

Read a plotted coordinate pair

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q22 Q22

7 marks

- 22 $ABCD$ is a kite, with diagonals AC and BD , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point M with coordinates $(0, 2)$

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,
find the coordinates of the points B and D .

(..... ,)

(..... ,)

WORKING SPACE

4MA1-2H Q22 Q22

7 marks

- 22 $ABCD$ is a kite, with diagonals AC and BD , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

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(..... ,)

(..... ,)

Chapter: Coordinate Geometry

7 marks

Q22 – Chapter: Coordinate Geometry

Family: Determine coordinates from geometric information • Variant: Kite or symmetric vertex coordinates

Method

The gradient of AM is $(2 - 4)/(0 - (-3)) = -2/3$.

Why: Use the two given coordinates.

Method

The kite diagonals are perpendicular, so BD has gradient $3/2$ and equation $y = (3/2)x + 2$.

Why: Use the negative reciprocal and point M.

Method

For B or D, $(x + 3)^2 + (y - 4)^2 = 6.5^2$.

Why: Each point is 6.5 units from A.

Method

Substitute $y = (3/2)x + 2$: $(x + 3)^2 + ((3/2)x - 2)^2 = 42.25$.

Why: Combine the line and distance conditions.

Method

Expanding gives $3.25x^2 + 13 = 42.25$.

Why: Expand and collect; the linear x terms cancel.

Method

Thus $x^2 = 9$, so $x = 3$ or $x = -3$; the corresponding y -values are 6.5 and -2.5.

Why: Solve the quadratic and substitute into the line.

Method

Since B has positive x -coordinate, $B = (3, 6.5)$ and $D = (-3, -2.5)$.

Why: Use the final labelling condition.

Final answer

B: $B = (3, 6.5)$

Final answer

D: $D = (-3, -2.5)$

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

H is the point with coordinates $(-4, 1)$

J is the point with coordinates $(j, 15)$ where $j < 0$

K is the point with coordinates $(6, k)$

M is the midpoint of JK .

The gradient of HM is 2

Find the value of j and the value of k .

$j = \dots\dots\dots$

$k = \dots\dots\dots$

Worked solution

January 2020 | Question 22 | 6 marks

Family: Determine coordinates from geometric information • Variant: Coordinates from isosceles and gradient constraints

Method / working

1. $M = ((j+6)/2, (15+k)/2)$.
2. Gradient $HM=2$ gives $(k+13)/(j+14)=2$, so $k=2j+15$.
3. $JK^2=80$ gives $(6-j)^2+(k-15)^2=80$.
4. Substitute k : $5j^2-12j-44=0$.
5. The condition $j<0$ selects $j=-2$.
6. Then $k=2(-2)+15=11$.

Final answer: $j=-2$, $k=11$

18 A rectangle $ABCD$ is to be drawn on a centimetre grid such that

A has coordinates $(-4, -2)$

B has coordinates $(1, 10)$

C has coordinates $(19, a)$

D has coordinates (b, c)

(a) Work out the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

..... cm

(3)

Worked solution

January 2021 | Question 18 | 4 marks

Family: Determine coordinates from geometric information • Variant: Rectangle or parallelogram missing vertex

Method / working

1. $AB=(12,5)$, so $b=14$ from the rectangle translation.
2. Gradient $AB=12/5$, hence gradient $BC=-5/12$.
3. Use $C=(19,a)$ to obtain $a=2.5$ and then $D=(14,-9.5)$.
4. State all three coordinates values.

Final answer: $a=2.5$, $b=14$, $c=-9.5$

23 $ABCD$ is a kite.

$$AB = AD \text{ and } CB = CD$$

The point B has coordinates $(k, 1)$ where k is a negative constant.

The point D has coordinates $(8, 7)$

The straight line L passes through the points B and D

The straight line L is parallel to the line with equation $5y - 3x = 6$

Find an equation of AC

Give your answer in the form $px + qy = r$ where p , q and r are integers.

Show your working clearly.



(Total for Question 23 is 6 marks)

Answer reference | January 2023 | Question 23

Family: Read and plot Cartesian coordinates • Variant: Read a plotted coordinate pair

23	$y = \frac{3}{5}x \left(+ \frac{6}{5} \right)$ or $y = 0.6x(+1.2)$ or (gradient =) $\frac{3}{5}$ or 0.6 oe		6	M1 for correct gradient which may be seen in an equation. Condone $\frac{3}{5}x$ or $0.6x$
	$k = -2$			B1 for $k = -2$
	$\left(\frac{-2+8}{2}, \frac{1+7}{2} \right)$ oe or (3, 4)			M1 for finding the midpoint (use of their k where $k < 0$)
	$\frac{3}{5}m = -1$ or $(m =) -\frac{5}{3}$			M1ft their gradient for use of $m_1 \times m_2 = -1$ Allow $-\frac{5}{3} = -1.67$ or better
	$"4" = -\frac{5}{3}"x"3" + c$ or $c = 9$ or $y - "4" = -\frac{5}{3}"(x - "3")$			M1 dep on M3
	Working required	$5x + 3y = 27$		A1 allow equation in any form where p, q and r are integers
Total 6 marks				

Worked solution

January 2023 | Question 23 | 6 marks

Family: Read and plot Cartesian coordinates • Variant: Read a plotted coordinate pair

Official final mark-scheme pages are embedded immediately after this question.